

Please refer to **Appendix A** pp **A2-A6** of the recommended text book.

1. UNIT 1.1

NUMBERS, INTERVALS AND INEQUALITIES

NUMBERS

(please refer to page A2-A3)

Integers ($\mathbb{Z}$): $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$

Rational Numbers ($\mathbb{Q}$): are **ratios of integers** or sets of numbers of the form $\frac{m}{n}$ where m and n are integers and $n \neq 0$.

- Examples: $\frac{1}{2}$, $\frac{3}{1}$, $\frac{1}{3}$, etc
- Numbers which can not be written as ratios are not rational numbers, and they are called **irrational numbers**. For example $\sqrt{2}$, $\sqrt[3]{7}$, $\sin 5$, π , e , $\dots$
- the corresponding decimal representation of rational numbers is repeating but not of irrational numbers. Examples

$$\frac{1}{2} = 0.5000\dots =$$

$$\frac{1}{3} = 0.3333\dots =$$

$$\frac{157}{495} = 0.317171717\dots =$$

$$\sqrt{3} =$$

$$\sqrt[3]{9} =$$

$$\pi =$$

$$e =$$

Numbers (Real numbers ($\mathbb{R}$)) : When we say numbers we are referring to the set of real numbers which is the union of the set of rational numbers and the set of irrational numbers.

The number line: A number line is used to represent real numbers. Every point on a number line corresponds to some real number.

Set: Set is a collection of well defined objects, and these objects are called elements of the set.

Set description:

- Complete listing method
- Set builder method

Union and intersection of sets: (carefully observe how we use the connectives *and* and *or*.)

Union : ($\cup$)

Intersection : ($\cap$)

The empty set: ($\emptyset$)

INTERVALS

Types of intervals and their geometric representation. Let $a, b \in \mathbb{R}$ such that $a < b$.

Open interval :

Closed interval :

Half open or Half closed interval :

infinite interval :

Combining intervals using $\cup$ and $\cap$.

Find the union and intersection of the following intervals and put your final answer in interval form

(1) $(1, 3), (-2, 2)$

$$(2) \quad [-8, 3), [-2, 10]$$

$$(3) \quad (1, \infty), (-\infty, -5]$$

INEQUALITIES

Rules of inequalities. (Please refer to page A4)

Examples: Use the rules of inequalities to solve the following and write your answers in terms of intervals

$$(1) \quad 2x + 7 > 3$$

$$(2) \quad 1 + 5x > 5 - 3x$$

$$(3) \quad 1 < 3x + 4 \leq 16$$

$$(4) \quad (x + 1)(x - 2) \geq 0$$

$$(5) \ x^2 + x < 2$$

$$(6) \ \frac{x+1}{x+2} \leq 0$$

$$(7) \ \frac{1}{x+1} > 1$$

$$(8) \ x^2 > 4$$

$$(9) \ x^2 < 4$$

$$(10) \quad -x^2 - x - 1 < 0$$